

# System Design Specification (SDS)

**Project:** Web Crawl, Accessibility & Compliance Monitoring System

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**Version:** v1.0

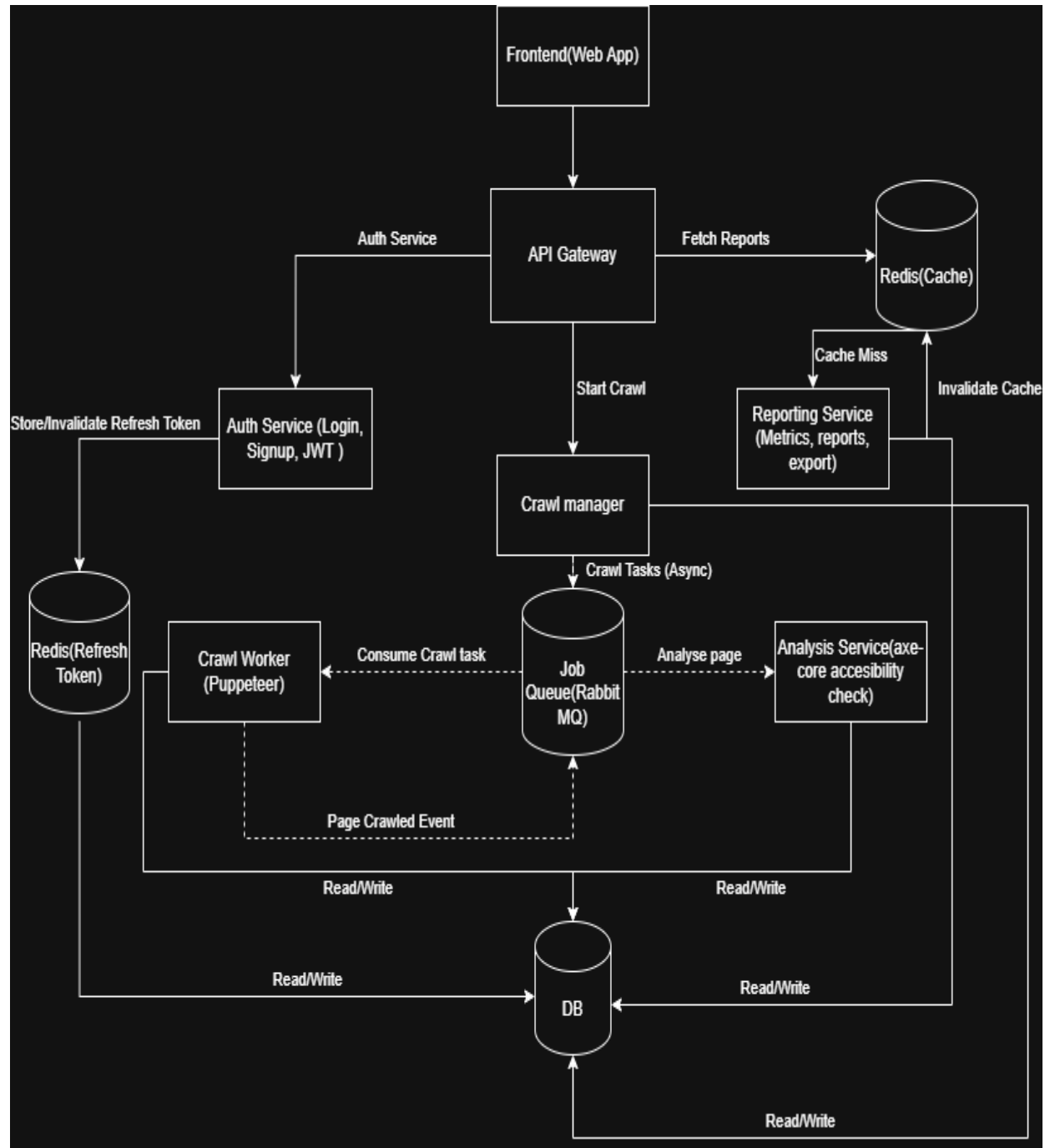
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**Status:** Draft

## 1. TECH STACK

| Layer                  | Technology                 | Why                                                             |
|------------------------|----------------------------|-----------------------------------------------------------------|
| Frontend               | React.js                   | To build UI for the app                                         |
| Backend                | Express.js with TypeScript | To create APIs on top of Node.js                                |
| Database               | PostgreSQL                 | To store data of various services and users in the application. |
| Caching                | Redis                      | To reduce the load on database and servers                      |
| Web Crawling           | Puppeteer                  | To fetch the DOM information of the specified URL.              |
| Job Queue              | RabbitMQ                   | For asynchronous communication between services                 |
| Accessibility Analysis | Axe-core                   | For accessibility testing of websites.                          |

## 2. SYSTEM ARCHITECTURE



## Database Design

